

MockMate: Leveraging AI for Adaptive and Realistic Interview Practice

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Abstract : The integration of artificial intelligence(AI) into career preparation tools is reshaping the way candidates train for interviews, offering scalable, tailored, and cost-effective solutions. MockMate is an AI-based interview chatbot to provide a realistic and individualized mock interview experience, and it also aims at improving both technical skills and accessibility. Mock mate reads and extracts the text from the resume using Natural Language Processing(NLP) and create questions based on their job role from five main software domains. It also gives small logic-based pseudocode questions to improve problem-solving. To check performance in real time, it looks the facial expressions using Convolutional Neural Network(CNN) and listens to the voice to find confidence using a Recurrent neural network(RNN), and also gives helpful feedback, avoiding repetition of the questions so the user can keep learning and get better at interviews

Keywords — Mock Interview, Facial Emotion Recognition, Voice Confidence Analysis, Problem-Solving Skills, Interview Training.

I. INTRODUCTION

In Today's world, many job seekers struggle with preparing interviews due to weak communication, confidence and find it difficult to customize their response to specific job roles. Conventional methods are often generic and fail to satisfy specific needs. In order to fill this gap, Mock Mate is an AI-Powered Mock interview that offers an engaging, customized and authentic interview experience. Utilizing Natural Language Processing, it evaluates a candidate's resume to determine their qualifications, experience, and professional goals before creating pertinent, role- specific questions for roles like Data Analyst, Software Developer, Data Scientist, Web Developer, and AI Engineer. In Order to guarantee a comprehensive assessment, Mock Mate leverages Deep Learning Models like Convolutional Neural Networks (CNN) to evaluate facial expressions, Recurrent Neural Network (RNN) are used to examine speech tone, emotional cues, and confidence levels throughout

responses to ensure a comprehensive review. During the interview process, a Conversational chatbot guides candidates with personalized questions, helpful guidance, and carefully chosen educational materials. Each session ends with a comprehensive scorecard that addresses technical accuracy, emotional delivery, and communication effectiveness. Mock Mate offers an effective tool for career advancement and interview preparation since it additionally suggests relevant job opportunities based on the candidate's profile.

II. LITERATURE SURVEY

This article presents the question and answering mock interview bot a computer program that is an AI-driven platform generating tailored questions from user resume[1]. This article introduces a virtual mock interview platform that lowers customer acquisition costs in the edtech industry by using AI technologies such as ADA2 for embeddings, NLP and GPT-4[2]. This work closes the gap between practice and real interviews by assessing the candidates on emotions through CNN for facial expression, confidence through speech recognition and knowledge[3]. This research presents a mock interview simulator that improves job preparation by employs speech recognition, text-to-speech synthesis and GPT-3.5 Turbo for customized and dynamic interactions[4]. This paper introduces a mock interview platform for virtual recruitment process using AI, with real time feedback and recent questions and this platform combines visual, audio and text based analysis for evaluation[5]. The study converts candidates responses into text and evaluates their relevance to the conversation flow[6]. This paper detects the incomplete answers and allows users to ask questions[7]. This paper introduces a VR-Based application for AI chatbot and support multiple languages[8]. This paper provides personalized AI mock interview with enhanced performance and gives feedbacks on communication[9]. This study analyzes the facial expression with audio to text and evaluates communication skills and proficiency using deep learning model[10]. This paper introduces AI driven mock platform that helps users to refine their skills and provides feedbacks and enhances

interview preparation[11]. This paper explores a small communication robot specialized for face to face interview communication practice[12].The AI mock interview platform assesses personality trait using NLP, sentiment analysis and CNN for confidence prediction[13]. This report investigates biases in software engineering interviews that affect users based on gender, race and some other factors[14].This paper explores five non invasive methods for assessing credibility and collects data during crime interviews and evaluate these methods[15]. This study explores robot mediated job interview training for the candidates and targeting nonverbal and physical communication skills[16]. This report presents VR based job application that is the combination of chatbot and virtual reality designed for software engineers to prepare for interviews[17]. This paper introduces a mock interview robot that analyzes the candidate's willingness to speak and answers accordingly to the questions[18].This paper explores the purpose of an AI driven system that analyze the performance and provides detailed feedback[19].This paper introduces an AI mock interview that evaluates user's performance ,speech, body language and enhance their efficiency and accuracy[20].

III. EXISTING WORK

A mock interview tool that creates personalized interview questions based on a job description to assist job seekers in efficiently preparing for interviews. It helps users polish their solutions, increase clarity, and boost confidence by giving them immediate AI feedback. Users can receive organized coaching based on best interview methods, practice responding to behavioural and technical questions via text or audio, and monitor their progress over time. With the help of this tool, people may improve their communication abilities, become experts at interviewing, and be ready for every situation that may arise during an interview.

A. Mock Star

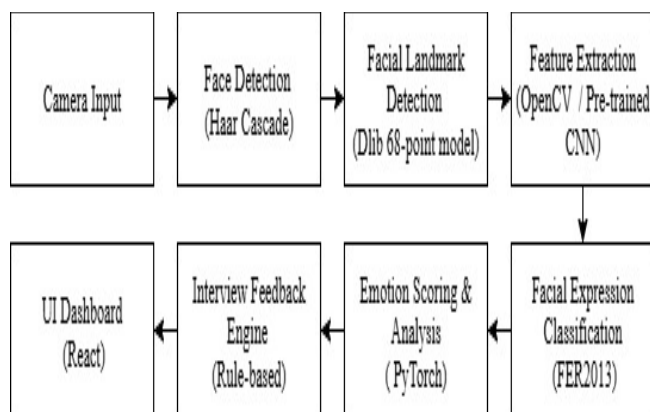


Fig 1. Workflow of Mock Star

Fig 1 represents that the system is an AI – powered mock interview platform, that offers coaching-style practice through live voice or video sessions. Using AI Avatars, it performs realistic and adaptive interviews based on the candidate's resume and job description. It helps to improve their verbal and nonverbal communication skills for real interviews. It adapts questions based on the performance and gives a comprehensive feedback report

that is provided at the completion of each session. This report examines their body language, eye contact, tempo, voice and answer structure.

B. Remasto

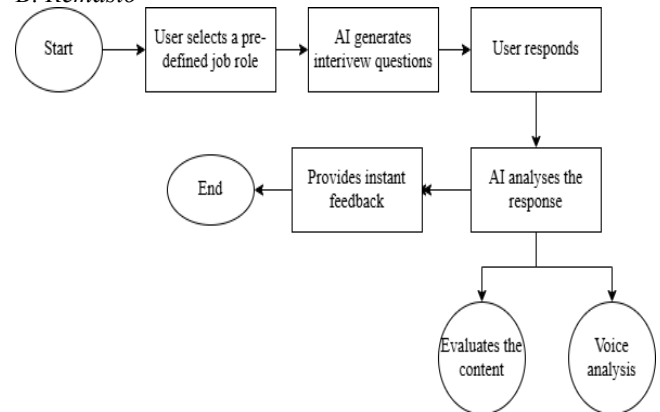


Fig 2. Workflow of Remasto

Fig.2 shows that the development of an individualized interview practice session, users start with the selection of pre-defined job profiles. The application mimics interview questions that range from technical to behavioral in nature. Feedback based on answers is provided to the users, and these answers can be stored as audio and text. The process of monitoring is used to recognize strengths and areas of improvement, enhancing self-confidence and communication abilities. Through ongoing practice, the system increases technical competence and responsiveness to varying interview formats, equipping users to excel in actual interviews.

C. Tools Comparison

TABLE I Comparison of existing tools

Features	Mock Star	Remasto
Model used	Uses NLP and speech analysis models for response understanding and scoring.	Utilizes ML and NLP models for resume parsing and interview question generation.
Resume Integration	Uses uploaded resume to customize questions and generate targeted feedback.	Deep integration of resume and company data for generating realistic, job-specific interviews.
Performance	Provides moderate real-time feedback with good response understanding; best suited for general interview practice.	Offers higher role-specific accuracy and realistic question context; best suited for company-based preparation.

Table I represents that both MockStar and Remasto make use of AI-based models for question generation, resume integration, and feedback, a number of limitations still persist that curb the possibility of them providing a comprehensive mock interview experience. Neither system uses multimodal analysis: assessing the

candidate's facial expressions, voice tone, or emotional state are important indicators of confidence and communication effectiveness. The feedback mechanisms in both platforms are mostly textual or speech content-based, lacking real-time behavioral or sentiment-based assessment. Third, their contextual adaptability is limited; follow-up questions are not dynamically adjusted by the systems based on the trends in candidate performance or interview flow.

IV. PROPOSED WORK

The proposed system is an artificial intelligence-powered mock interview platform integrated with multimodal analysis, which assesses and improves the technical and confidence level of the candidates. It predicts suitable job roles from uploaded resumes using NLP and generates domain-specific, non-repetitive pseudocode questions. This system scores live responses using CNN-based facial emotion detection and RNN-based voice confidence scoring, providing adaptive feedback and personalized learning resources to help in continuous improvement.

A. System Architecture

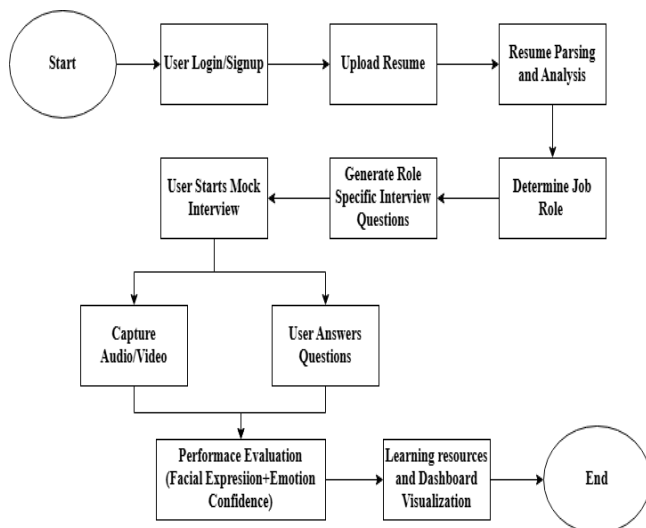


Fig 3. Architecture of Mock-Mate

Fig3. Represent the architecture and working of the system, in that the system first it allowing the user to upload their resume in PDF or DOC format. Once uploaded, the text is extracted using Optical Character Recognition (OCR) or PDF parsing techniques to ensure accurate retrieval of the resume content. The extracted text is then processed for keyword extraction, where techniques like TF-IDF and Named Entity Recognition (NER) are applied to identify important details such as skills, educational qualifications, and work experiences. Based on this structured information, the system performs role prediction by matching the extracted data with predefined criteria to classify the candidate into one of the five supported software roles. Once the User role is predicted the system retrieves a set of role specific questions from web. The questions set is typically includes technical theory questions that candidates understanding of core concepts relevant to the predicted role.and also give the pseudocode challenges to improve the logical thinking and problem solving ability of the candidate.to maintain the uniqueness and quality of the interview experience the system integrates a session tracker

that records all previously asked questions for the user.this system avoids the repetition and ensure that each session prevents fresh and diverse content.In result user continuously challenged with new scenarios, allowing for a more accurate assessment of their evolving technical skills Facial emotion detection(CNN) uses a convolutional neural network to analyze the video input, and identifies the emotions such as confidence, nervousness.it provides emotional insight into the candidate's responses.and also give the confidence score using RNN to analyse the audio features like tone, fluency and pitch variations and also give the confidence score based on the confidence level.MockMate provides the score based on their performance in technical, behavioral, emotional and vocal aspects. learning resources, recommends role based tutorials, coding practice platforms and communication skill guides.suggests the suitable jobs based on role prediction and skill set. Adjust future interview questions difficulty and focus areas based on previous performance.

V. MODEL DEVELOPMENT

A. CNN Model Development

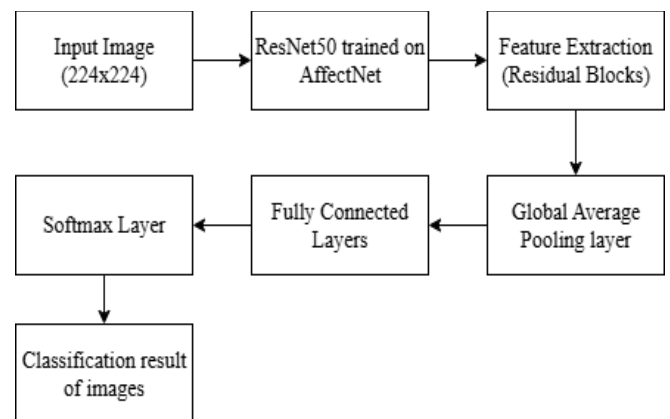


Fig 4. Architecture of ResNet50 model

Fig4. represents that the MockMate system has been evolved to use with the ResNet50 CNN model pre trained on a AffectNet dataset to recognize the facial emotions in real time. The AffectNet dataset consist of more than a million of face photos from online searches with 1250 emotion related keywords. The dataset has eight emotion categories such as angry, disguised, fearful, glad, sad, surprised, neutral and contempt with their annotated labels for 450000 images. It has a strong representation that generalizes across various ethnicities, lighting conditions, positions and occlusions to generalize the data in diverse situations and environments. The ResNet50 model has base feature extractor by adjusted on the FER+ dataset to fit the MockMate's target categories for interview-style with grayscale settings. With its deep 50 convolutional layers, the ResNet50 model was trained efficiently to predict the emotions and also reduce the vanishing gradient problems during prediction. During inference, the candidate's frames are normalized and scaled to 224x224 pixels to meet the input requirement of ResNet50 model and then processed by Haar cascade for face detection. The ResNet50 model get these processed frames to extract the deeper facial traits to analyse the emotion of the candidate. Lastly a fully connected convolutional layer will collect these processed data from average pooling layer and produces probability by

using softmax classifier across the emotion classes. The softmax classifier is:

$$\sigma(z)_i = \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}}$$

where z_i is the logit score for class i and k is the number of emotion categories. The model is trained using the Adam optimizer with a learning rate of 1×10^{-4} for efficient weight updation of the model to gradually maintains the Affectnet feature representations. The softmax classifier outputs the probability distribution over emotion classes like {happy:0.78,neutral:0.15,nervous:0.07}, thereby predicting the candidate's emotion state.

TABLE II CNN Model Summary

Layer Type	Number of Layers	Components Used
Convolution Layer	3	Filters(32,64,128), Kernel size(3x3)
Batch Normalization	3	Applied each convolution layer
Max Pooling layer	3	Pool size(2x2)
Dropout Layer	2	Dropout rate 0.5
Dense layer	1	Fully Connected(128 neurons,ReLU)
Softmax Output Layer	1	Number of Classes = 8

Table II represents the architecture of convolutional neural network which has been used for multi class classification in Mockmate system. The network is made up of three convolutional layers where each layer is followed by a batch normalization to stabilize the learning. Then to form a greater generalization there were two dropout layers were used randomly drop out some layers and the max pooling layer of size 2x2 used to prevent the model to perform from overfitting. Lastly, the softmax layer is used to classify the target classes which is optimized to balance the efficiency of feature extraction.

B. RNN Model Development

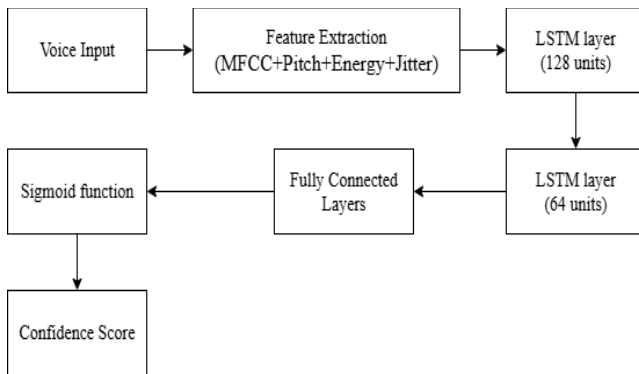


Fig 5. Architecture of RNN model

Fig5. represents the MockMate system integrates a Recurrent Neural Network with Long Short Term Memory(LSTM) layers to evaluate the candidate's confidence and vocal delivery during the interview simulations. The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDEES), which contains of 1440 high quality speech sample of 24 directories where the 12 directories having the male voice and the another 12

directories having the female voices is used to train the model. These samples covers eight different emotion classes like neutral, calm, happy, sad, angry, fearful, disgust and surprise. This dataset helps the model to learn a variety of vocal expressions to generalize on different situations which is crucial for reliable confidence estimate. The audio preprocessing pipeline includes noise reduction, amplitude normalization and frame segmentation processes. From each frame of voice, the model extracts the Mel-Frequency Cepstral Coefficient(MFCCs) along with its prosodic features such as pitch, energy and jitter. Pitch is computed as:

$$F_0 = \frac{1}{T(\sum_{i=1}^T)} * f_t$$

where, f_t is the fundamental frequency at time t over a time period of T .

$$I = 10 \log_{10} \left(\frac{1}{T(\sum_{i=1}^T)} \right) * (x_t)^2$$

where, I measures the energy of speech signal and x_t represents the amplitude of the speech signal.

TABLE III RNN Model Summary

Layer Type	Number of Layers	Components Used
LSTM Layer	2	Units(128,64)
Dropout Layer	2	Dropout rate 0.3
Dense layer	1	Fully Connected(64 neurons, ReLU)
Sigmoid Output Layer	1	Confidence score in [0,1]

Table III shows the architecture of the Long-Short Term Memory which is used for temporal feature extraction and confidence prediction. The model composed of two LSTM layers one for capture long-term dependencies and sequential relationship from input data. The model terminates with a sigmoid layer that produces the probability of the targeted outcome between 0 and 1.

C. Model Performance

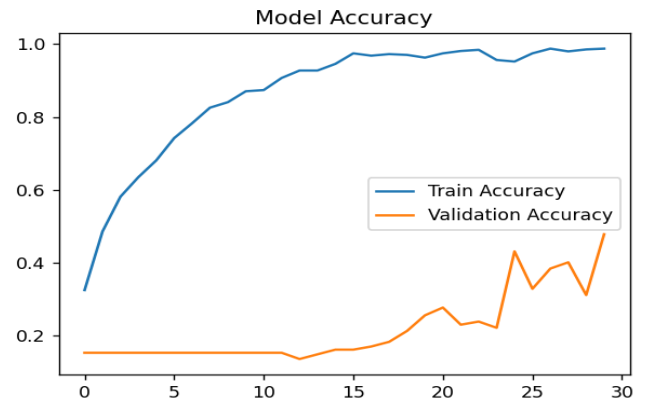


Fig 6. Accuracy of the model

Fig 6 represents the model's performance is evaluated through train-test accuracy assessing its ability to generalize on unseen data. Validation metrics such as accuracy, loss, validation accuracy and validation loss were performed to

reduce the overfitting. Prosody evaluation plays a crucial role in analyzing between features like pitch, rhythm and tones to distinguish between various emotions.

VI. RESULT AND DISCUSSION

A. Dataset description

The RAVDEES dataset is a publicly accessible dataset containing voice records. It has total of 24 files where 12 files are male voice recordings and 12 files are female voice recordings. The recorded voices are in different emotions such as calm, angry, stressed, happy, fearful, neutral and sad. The recordings are in the format of .wav files denoted as (03-01-XX-XX-XX-XX-XX.wav) the third xx segment denotes the emotion of the voice.

B. Accuracy and precision analysis

TABLE IV Accuracy and Precision Table

Products	Accuracy	Precision
Mock Star	85%	81%
Interview Boss.AI	88%	84%
Proposed	97%	90%

Table IV is a comparative analysis of the suggested AI-powered interview system with current solutions, Mock Star and Interview Boss.AI, based on accuracy and precision. Accuracy is a measure of the system's overall accuracy in making predictions about interview results, and precision is a measure of the system's precision in identifying relevant answers without generating false alarms.

Based on the results, the suggested system has an accuracy of 97%, which is much more than Mock Star (85%) and Interview Boss.AI (88%). Likewise, the precision of the suggested system is 90%, which is better than Mock Star (81%) and Interview Boss.AI (84%). These developments suggest that the suggested method is more consistent in accurately assessing candidate responses and reducing errors, exhibiting its better performance in developing an effective and adaptive interview test setting.

C. Results

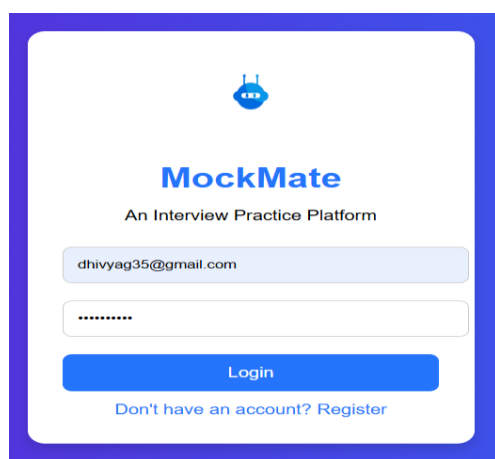


Fig 7. Login page of MockMate

Fig7. shows the login page of mockmate system, which is the point of entry for users to access the system. Users need to log in through their credentials to maintain a personalized interview sessions for each user. This process

act as a basics for user-specific interview preparation and tracking.



Fig 8. Job Role Prediction

Fig8. represents the domain selection window of the Mockmate system which helps to tailor the questions based on the job role predicted. Once the domain was predicted the system utilizes the online resources to scrape a set of interview questions that represent the competencies, technical skills, and industry expectation questions. This way the users can tailored through domain specific questions rather than generic questions. By adjusting the interview material to specific career objectives and level of skills, this feature maximizes the effectiveness of the practice session, assisting users in boosting confidence, refining subject matter knowledge and strategically preparing for competitive job interviews.

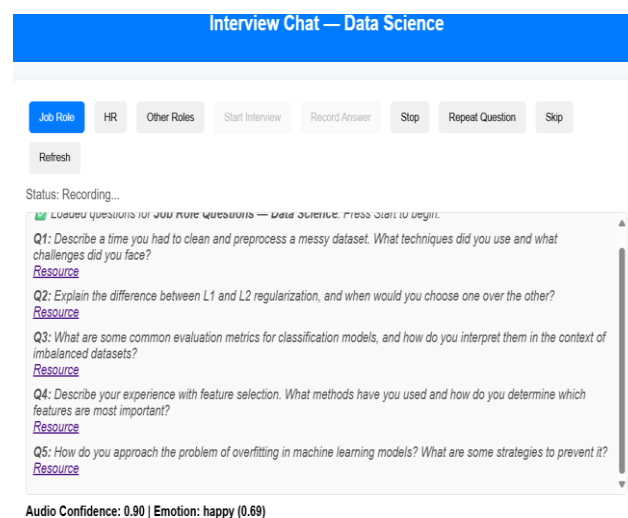


Fig 9. Webpage of MockMate

Fig 9 depicts the MockMate interview panel's functional interface with five fundamental operation controls that aim to provide user engagement and allow AI-based assessment. The Read button uses Google Text-to-Speech to vocalize questions, making it more accessible and clearer to the user. The Record button starts capturing the user's oral response, whereas the Stop button stops recording to allow further

processing. The Play option enables users to listen to their recorded response themselves for self-evaluation, and the Submit option sends the recorded audio to the inbuilt deep learning models for analysis. For comprehensive assessment of the candidate's potential, Convolutional Neural Networks (CNNs) are employed for facial expression recognition, while Recurrent Neural Networks (RNNs) examine speech tone, voice patterns, and emotional traits. This multimodal assessment model improves the validity and trustworthiness of feedback, presenting users with thorough evidence of their interview preparedness.

VII. CONCLUSION

This AI-powered mock interview system offers an intelligent platform for effective interview preparation through dynamic question retrieval, real-time evaluation, and AI-driven feedback. While it enhances user confidence and simulates realistic interview scenarios, the challenges remain in capturing non-verbal cues, ensuring bias-free assessments, and improving contextual adaptability. Further development could be done on multimodal emotion analysis, AI-based resume evaluation, and hybrid AI-human feedback to make assessments more accurate and personalized. Expanding this system across diverse industries and job roles, with adaptive learning enhancements, will further strengthen its effectiveness and real-world applicability.

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